

SPID Antenna rotator SETUP / Getting Started V 12.1

The instructions presented here should suffice to install or to use the basic functions of the rotator and controller.

Installation Considerations

Mechanical installation to your mast and/or mounting plate is straightforward but there are a couple of concerns if mounting to a very thin mast or a very hard mast. The 8 bolts exert a considerable force on the mast and will tend to dent a soft or thin mast if over tightened. You might wish to consider inserting a sleeve of suitable diameter to reinforce the section of mast in the rotator. For very hard mast material that won't dent,, it may be useful to make a small indentation with a drill bit to allow the bolts to grip better. However for most normal Aluminum or steel masts it will work fine without any special precautions or modifications.

If turning a heavy antenna, a longer mast either above or below the rotator (or both), will act as a shock absorber. This is a good thing. Be sure to allow enough slack in the coax loop to allow for full rotation and to take advantage of the 180 degrees of over travel on both sides.

Waterproofing

Before putting back the plastic cover over the terminal strip, a thin bead of silicone or other caulking compound around the edge will prevent any leakage there. For very windy wet locations a rubberized weatherproof rain hood might be beneficial. Some users have reported that driveway marker cones work for this purpose. If you live in a location where you get "horizontal rain" you may wish to caulk the underside of the top metal housing covering the top shaft.

Where's North?

Well, you don't really care right now. Put up the rotator and tighten the mast bolts. Once on the ground then simply use the controller to turn the antenna to north. Don't worry what the readout says or if it even rotates in the right direction. Then zero the controller by turning it off, then holding the "F" button while turning the power back on. Now check to see that it rotates in the proper direction in accordance with the right and left arrows on the front panel. If not, simply reverse the 2 wires going to the motor. That's all the calibration needed.

Please also read: "SETUP MENU", this would be very help full to getting started !

Power supply

All SPID standard rotators can handle **12 Volts up to MAX 18 Volts DC for all SPID Standard Antenna Rotors !**

If needed max rotation torque, it's recommend to use a 18 Volts power supply.
RF HAMDESIGN has available several power supplies.

**For all SPID Standard rotators:
MAX OPERATION VOLTAGE = 18Volts DC**

Connect rotator at control unit

The connections to the motor and direction sensing circuits are under the plastic cover on the side of the motor. (AZ&EL type rotator has a connections possibility in the centre of the rotator) Remove the screws holding the plastic weather cover for access to the terminal strip. It is suggested to use a short pigtail (several feet long) of smaller gauge wire that will more easily fit under the plastic weather cover, and then connect this to your longer run of heavier cable. This pigtail is a convenient way to connect to the controller to test the operation.

Polarity: The wires to the motor are polarity sensitive as they determine the direction of rotation, but it is generally easier to reverse them at the controller end if the motor turns in the wrong direction. The two direction sensing wires have no polarity concerns. The power for the controller is automatically derived from any polarity DC or even AC voltage so you may connect the fixed lead from the controller (typically brown and blue wires) either way to your 12 to 18 volt (approx 10 amp) power supply. A small battery (12v gel cell type) capable of several amps is quite adequate for testing purposes and can be used for initial positioning of the antenna.

Connector connection controller and rotator: number 1 and 2 motor rotator (12 – 18 Volts DC), Number 3 and 4 puls detection, AZ&EL rotator same but add number 5 and 6 motor rotator and 7 and 8 pulls detection.

Controller switch on / off

The best way is to switch off the rotator controller first when used as tracking rotator and when it is connected to a PC. After this you are ready to close the PC application.

!!! VERY IMPORTANT: !!!

MAX OPERATION VOLTAGE = 18Volts DC

SETUP MENU

For operation it is recommended to go to the Setup Menu by pressing the "S" button to **configure** your controller, each press @ "S" you will walk through the Setup menu;
(Note: AZ rotator **has no** mechanical stop, **EL rotator has a mechanical stop**)

Good value's for a good rotator rotation is:

(Change value's with up/down and left/right buttons)

Azimuth model: (press < > buttons to change value & next step press S)

P or blank = 1 (BIG-RAK = 0.5) = Programmable Rotor gear value. The first setting should be 1.0 or 0.5 , it is possible to have one more setting depending on the version you have. Press the arrow keys to adjust. Otherwise the default setting is 1.0 and the unit only moves half distance...(0.5 degree model)

PH= 10.0 with blink-led

PL= AZ= 350.0 + fixed led (No blinking led !)

P(0)= Programmable Reset Value (0 or 180 degrees)

P1...P6= 6 free programmable preset buttons for use with mouse

PS= SP (SPID)

PP or blank = 0.0 / 0.0 (if offset is needed, see manual)

Press "F" button to go to normal operation or PC tracking option, ready.

Azimuth & Elevation model: (press < > buttons to change value & next step press S)

P or blank = 1.0 / 1.0 (BIG-RAS = 0.50 / 0.50)

PH= AZ part=10.0 with blink-led & EL part=90 with 1 led (if EL 180 dgr is needed set EL part =180)

PL= AZ= 350.0 + extra led & EL= -5.0 + extra led

PS= SP (SPID)

PP= 0.0 / 0.0 (if offset is needed, see manual)

Press "F" button to go to normal operation, ready.

Software and SPID :

To control the rotator by PC software there are several options:

We do recommend PST Rotor (http://www.qsl.net/yo3dmu/index_Page346.htm) Other option is ORBITRON (<http://www.stoff.pl/>) But this software needs more experience and is only for AZ & EL rotators.

You also have **to install a driver for Orbitron**, we do recommend: *Spidalfa 0.97 driver*, it's also at the ORBITRON web-site at tab: Download/DRIVERS.

Of course there are more applications available which can drive the SPID controllers, please pay attention when installing other drivers, always **install 1 at time !!**

Control SPID (SPX) Rotators through DDE-Cient

SPID and SPX Model Azimuth and Elevation Rotor work very well out-of-the-box with USB control box Rot2Prog from tracking software like for example WXtrack, Orbitron, Moonsked etc

When controlling this rotor from the popular Tracking software you have to use a DDE client (I fact is not build in already, for example PST rotor is plug and play).

These are the steps to setup up successful rotor control:

Installation of USB drivers for Rot2ProgIf you have problems with automatic installation of Windows virtual COM/USB-drivers for the Rot2Prog controller, download drivers from <http://www.ftdichip.com/Drivers/VCP.htm>

Wiring RAU, RAK or BIG-RAK SPID Rotator:

Wiring Connections:

The rotator unit must be wired to the control unit with 8 wires cable. 4 wires – azimuth (1,2,3,4) and 4 wires - elevation (5,6,7,8). Diameter of wires in cable to connect the control unit to the rotator depends upon the distance between rotator and controller. Control cable with 1.5mm² conductors will do fine up to 50 meter length.

You can enter at the site of the rotor by removing the metal plate by removing the 2 screws which hold this plate in place, you will find the terminal block to connect the wires. Numbering is from left to right.

Azimuth:

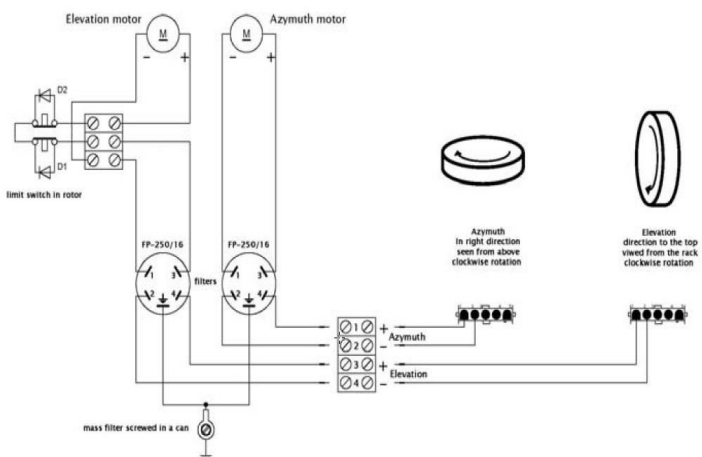
- ✓ 1 Motor Drive to Azimuth 1 on controller terminal
- ✓ 2 Motor Drive to Azimuth 2 on controller terminal
- ✓ 3 Impulse Sense to Azimuth 3 on controller terminal
- ✓ 4 Impulse Sense to Azimuth 4 on controller terminal

Elevation: *

- ✓ 5 Motor Drive to Elevation 1 on controller terminal
- ✓ 6 Motor Drive to Elevation 2 on controller terminal
- ✓ 7 Impulse Sense to Elevation 3 on controller terminal
- ✓ 8 Impulse Sense to Elevation 4 on controller terminal

Please note that the coax from the antenna be made with extra length to allow for the "Over Travel" !

* Note: The elevation has limit switch build in, pay some attention durign test setup, you must turn it back to zero !



Wiring for BIG-RAK, RAK, RAU or REAL SPID Rotator:

Wiring Connections:

The rotator unit must be wired to the control unit with 4-wire cable. The Diameter of the 4-wire cable to connect the control unit to the rotator depends upon the distance between rotator and controller. Control cable with 1.5mm² conductors will do fine up to 50 meter length.

Remove black cover from the motor body as shown and make connections as follows:

Rotor connection to controller:

- ✓ 1 Motor Drive to 1 on controller terminal
- ✓ 2 Motor Drive to 2 on controller terminal
- ✓ 3 Impulse Sense to 3 on controller terminal
- ✓ 4 Impulse Sense to 4 on controller terminal



Wiring for RAS and BIG-RAS Rotator:

Wiring Connections:

The rotator unit must be wired to the control unit with 8-wire cable. The Diameter of the 8-wire cable to connect the control unit to the rotator depends upon the distance between rotator and controller. Control cable with 1.5mm² conductors will do fine up to 50 meter length.

Remove black metal cover plate, it is fixed by 2 screws, you can find this plate on the site of the rotator, you can now enter the junction place and you will find the 2pc connections blocks.

Make connections as follows:

Rotor connection to controller for Azimuth and Elevation

- ✓ 1 Motor Drive to 1 on controller terminal
- ✓ 2 Motor Drive to 2 on controller terminal
- ✓ 3 Impulse Sense to 3 on controller terminal
- ✓ 4 Impulse Sense to 4 on controller terminal



Important note: Do not accidentally switch the motor wires with the impulse wires. Damage to the control unit may occur!

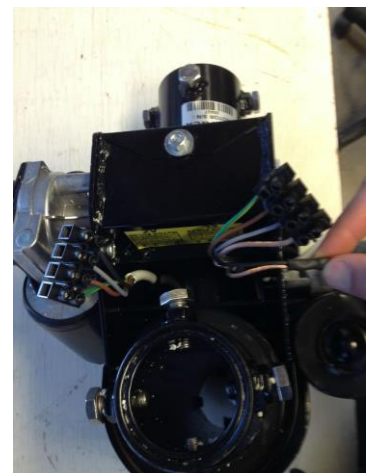
Tool List:

- Screwdriver (Phillips)
- Utility Knife
- Adjustable Wrench
- Small Screwdrivers (Phillips & Flathead)
- Soldering Iron Solder

Instructions:

On the rotator, there is a plate with two screws and two waterproofed wire outputs. On the face next to the plate should be a sticker detailing the EL AZ wiring.

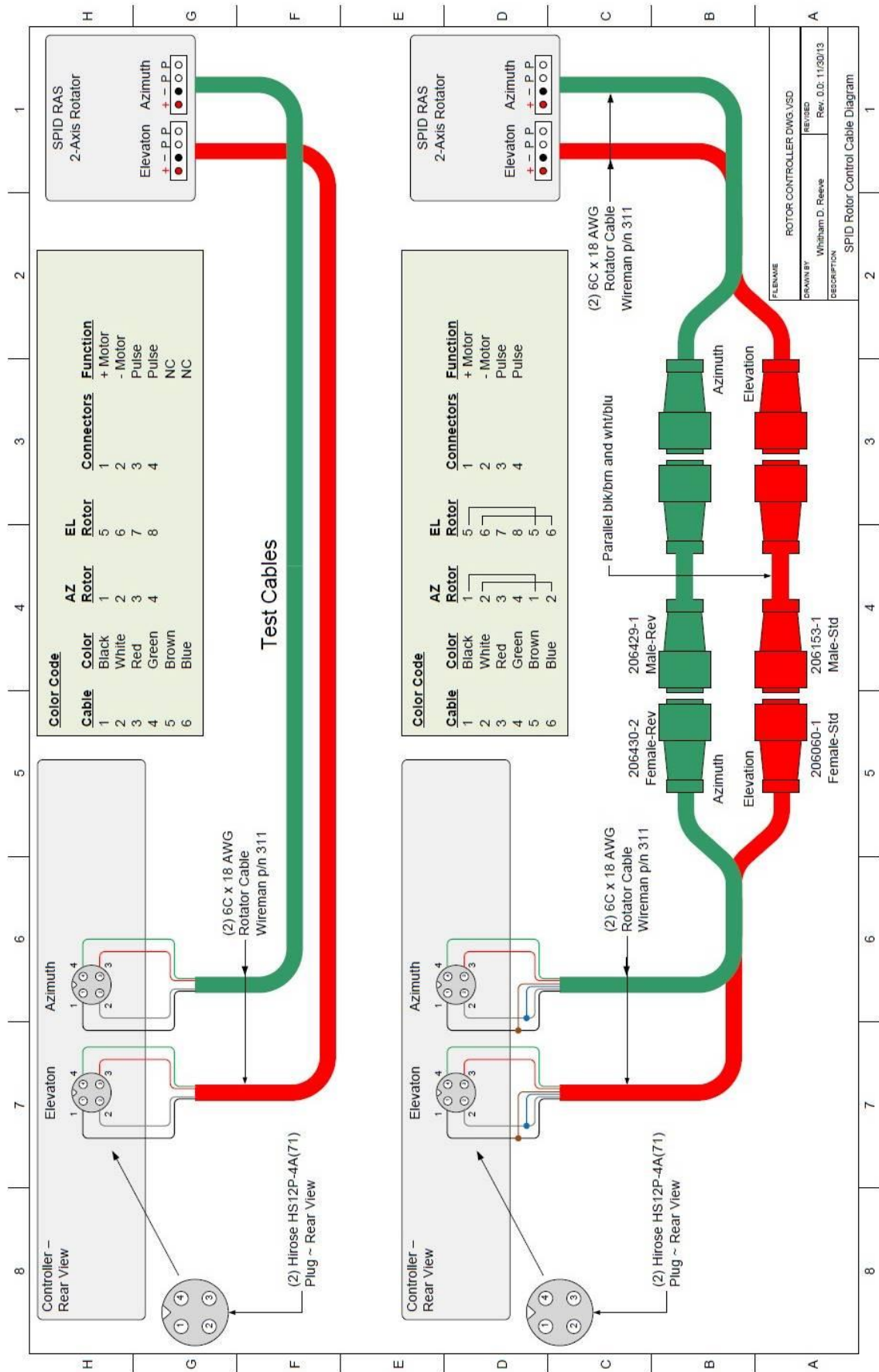
- a) First, unscrew/remove the two waterproof outlets, and unscrew the two screws on the plate.
- b) The edge of the plate may be painted over, so use a utility knife to edge the plate. Using the back of a screwdriver, tap around the corners of the plate to loosen it. Using a screwdriver or some other means of leverage, insert one end into a hole left from unscrewing the waterproof outlet, and pry off the plate. This may take some time since the plate has a tight fit.
- c) Underneath the plate are two wiring terminals, one for AZ and one for EL.
- d) Test Cables to Controller:....
Before you mount the rotator on a tower outside door we recommend to create a set of test cables, approx. 75 cm long. You can create 2 pcs of 4-core cables. The ends of the wires were stripped and tinned.
- e) The free end was then soldered to a head which connects to the controller.



Connection of a SPID Controller to a Rotor

Drawing is for a Dual Ax rotor controller (Rot2Prog).
For use with a single Ax rotor, 1 plug will be used (AZ or EL)

Second drawing is an example how to connect with extra connectors when you need fast disconnections.



Trouble shooting section

Rotators with Elevation rotation:

There are some things which are important to know:

Each rotor with a Elevation section has mechanical stops, if you test your rotor at the workbench and you do not turn it back to zero after you stop testing, you will have left not the full 180 degrees anymore to rotate when mounted.
This means, when you make tests, always turn back as far the rotor can to the zero degree mechanical stop.

When you rotor stops at, for example 40 degrees during operation, the rotor is at one of the mechanical stops

It is time to reset the Elevation rotor:

#1 To turn your rotor back to zero, remove rotor from controller, connect 12Volts DC to the wires of the rotor, (Pin 1 and 2 of the connector) Watch now if the rotor turns
If not, swap over + and - and try again.
Let your rotor turn until it stops, it is now at zero degree or 180 degree !

#2 When done, connect rotor to your controller again
Switch off the controller.
Switch the controller back on while pressing the F button, this will zero the controller.

3 It is time to check your settings, DO NOT ROTATE YOUR ROTOR !!!
Most used settings can be found on page 2 of this sheet.
Follow the Elevation settings.
It is IMPORTANT NOT TO TURN THE ROTATOR UNTIL THIS HAS BEEN COMPLETED.
Pay attention to the PH and PL values.
Once completed your rotor is ready to use.

Controller connected to PC:

When using the controller in emulation mode please ensure that:

- The baud rate is 1200, 8 data, 1 stop for the AZ units (Rot1Prog)
- And 600, 8 Data,1 Stop for AZ/EL units (Rot2Prog).
- The controller must be set to " A " mode with the " F " button
- The controller must be set to the emulation mode you are using ie. Yaesu

Maintenance by using WD-40 Spray:

Recommendation is to spray WD-40 into your AZ and/or AZ&EL motor direct.

There is a little hole into the motor which is used to drain moisture, in this hole you should spray 2 time a year some into the motor.

WD-40 is mostly a solvent with a very light lubricant mixed in. It's great for getting stuck parts moving again. When you spray it on, the solvent dislodges whatever gunk may be causing the part to stick and then evaporates, leaving a light lubricant behind. This will allow the previously stuck part to move again.

No worry to use this spray, you can spray as much as you like !

In the Red circle you can find the little hole, if there is not hole you can drill it on that +/- Place. (Not critical)



Mouse Controller:

The optional mouse controller allows easy desktop access to the most commonly used front panel controls. These buttons are functionally equivalent to the corresponding front panel controls.

Control buttons which are available:

- Left (Decrease)
- Up (Increase)
- Down (Decrease)
- Right (Increase)
- S = Setup mode or stop

